

Study & Test Marks — Calculation Rules

Test schedule: 3, 5, 7, 9, 12, 14, 16 May (tests) and 17 May (final test). Study window: 3–17 May.

1. Effective Study Hours

Setup. Each batch has 10 students who sit together in one room. On a **study day** a student studies in two sessions — half the day's allotted hours in the morning (starting 9:00–10:00 AM), a break, then the other half in the evening (starting 3:00–4:00 PM). On a **test day** the test is in the morning and the student studies the full allotted hours in a single evening session (starting 5:00–6:00 PM), after the test. There is no study on 17 May.

1.1 Effective batch study time (per session)

Because all 10 students share a room, a student's effective study time is the time during which the **whole batch** is studying together. For each session:

- **Start** = when the **last** student in the batch starts studying (only then is everyone studying).
- **End** = when the **first** student in the batch gets up (once one student leaves, not everyone is studying).

| *Effective batch session time = (time first student gets up) – (time all students were studying)*

This value is the **same for every student** in the batch.

1.2 Effective batch study time (per day)

- **Study day:** morning session overlap + evening session overlap.
- **Test day:** the single evening session overlap.

1.3 Student's effective study time (per day)

Each student also gets partial credit for time they personally studied beyond the batch overlap:

| *Student daily effective hours = daily batch effective + 0.2 × (student's hours that day – daily batch effective)*

In words: every student receives the full shared batch-overlap time for that day, plus 20% credit for the extra time they personally studied beyond it.

1.4 Totals

- **Effective study hours between two tests** = the evening study after the earlier test + the effective hours of the study days in between (a test is in the morning, study happens afterwards).
- **Total effective study hours (3–17 May)** = the sum of the student's daily effective hours over the whole period (17 May has no study).

2. Marks for Tests After 3 May (up to 16 May)

Base. The 3 May test is the base score (out of 360); everyone is present on 3 May. Each later test updates a student's marks using one of two cases, depending on whether the student was present in the **previous** test.

Case A — student was PRESENT in the last test

Marks this test = (previous test marks + 2 × effective study hours between the two tests) – (minutes late this test)

Case B — student was ABSENT in the last test

Marks this test = (last recorded marks + 2 × effective study hours between this test and the missed previous test) + (effective study hours between that missed test and the test before it) – (minutes late this test)

Notes

- Effective study hours used here are the **effective** hours from Section 1 (not the total allotted hours).
- A student who is **absent** in a test has no marks for it (shown as “Absent”); their last recorded marks carry forward to the next test they attend.
- Late minutes (0–4) apply only when the student is present; each late minute reduces the score by 1.
- Marks are out of 360 and can be fractional, because effective study is measured in hours.

3. Marks for the 17 May Final Test

On 17 May **everyone is present** (no absentees). The final-test marks are:

17 May marks = marks obtained in the last test the student was present in + total effective study hours from 3 May to 17 May

Notes

- **Marks in last present test** = the student’s most recent recorded marks up to the 16 May test (for a student absent in their last test, this carries forward from the previous test they attended).
- **Total effective study hours (3–17 May)** = the sum of the student’s daily effective study hours over the whole period (17 May itself has no study).
- Marks are out of 360.